

EL MEHDI HARESS

PhD in Applied Mathematics | CentraleSupélec engineer

Statistical modelling | Python | Machine learning | Shipping products

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PROFILE

Quantitative researcher with **5 years** of work on models that break under the usual assumptions: discretisation schemes with proven convergence, multi-parameter estimation from discrete data, sensitivity analysis, and **risk-aware learning** (AISTATS). Eight peer-reviewed papers. I write **Python** daily and ship tools other people use. I want that same rigour applied to **product data**: pipelines you can trust, metrics you can defend, and decisions rather than theorems.

SKILLS

Modelling	Stochastic models, Monte Carlo, parameter estimation, sensitivity and robustness, uncertainty quantification, first-principles problem solving.
Data & ML	Statistical inference on observed processes; spectral risk measures (VaR, CVaR); heavy tails; risk-constrained learning. Everyday scientific Python.
Engineering	Python (NumPy, SciPy, pandas) — 8 years; Git; MATLAB, R; JavaScript/TypeScript, React, Firebase. C++ basics. Open-source research code.
Working style	Clear writing and talks; teaching; pairing with non-specialists; owning a problem end to end. Comfortable with English as a working language.
Languages	French — native. English — fluent (2 years teaching and research in the UK). Japanese — beginner.

EXPERIENCE

09/2026 — now	Research engineer — probabilistic numerical methods Inria Sophia Antipolis — CALISTO (Mireille Bossy) ► Design and analyse numerical schemes for industrial multi-scale stochastic systems; implement in Python and check the claimed convergence rates.
12/2024 — 08/2026	Research fellow in stochastic analysis University of Leeds — K. Dareiotis and K. Lê ► Proved uniform pathwise stability for singular SDEs driven by fractional Brownian motion (<i>SPA</i>, 2026); quantified how solutions react to changes in model and parameters. ► Co-organised FRSSD 2026 (18 speakers). Taught stochastic processes (undergrad, 50 students).
10/2021 — 11/2024	PhD — numerical approximation of singular SDEs and SPDEs Université Paris-Saclay / CentraleSupélec — L. Goudenège and A. Richard ► Built a rigorous approximation framework where standard Euler assumptions fail (<i>IMA J. Numerical Analysis</i>, <i>SPA</i>). ► Joint estimators (drift, volatility, Hurst) from discrete observations; 5 public repositories. ► Taught PDEs, probability, and stochastic finance (M2) at CentraleSupélec.
02/2020 — 08/2020	Research intern — learning under risk constraints Osaka University — M. Holland ► Spectral risk objectives under unbounded losses and heavy tails; 2 <i>AISTATS</i> papers with implementations and experiments.

SELECTED PRODUCTS

Nerduncover	Multiplayer web game (Undercover-style, pop culture). Designed, built and shipped end to end.
Hana Bella	hanabellaspa.pro — live client site and admin back-office (accounts, roles, period reports, CSV export). React, TypeScript, Firebase.
Personal site	elmehdiharess.org — designed and maintained; React / TypeScript / Vite.

EDUCATION

2021–2024	PhD in Applied Mathematics — Université Paris-Saclay.
2017–2021	Engineering degree, Applied Mathematics — CentraleSupélec. M2 Mathematics of Randomness (2020–2021).
2015–2017	CPGE (MP) — Lycée Pierre de Fermat, Toulouse.

SELECTED PUBLICATIONS

8 papers in total; full list at elmehdiharess.org. Author order is alphabetical in probability.

- M. Holland, **E.M. Haress**. *Spectral risk-based learning using unbounded losses*. AISTATS, 2022.
- M. Holland, **E.M. Haress**. *Learning with risk-averse feedback under potentially heavy tails*. AISTATS, 2021.
- L. Goudenège, **E.M. Haress**, A. Richard. *Numerical approximation of SDEs with distributional drift*. SPA, 2024.
- K. Dareiotis, **E.M. Haress**, K. Lê. *Uniform pathwise stability of additive singular SDEs driven by fractional Brownian motion*. SPA, 2026.